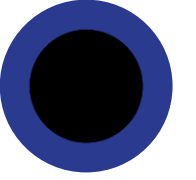




TECH2

Week 1 Lecture

Introduction to Programming, Data, and Information Technology



Welcome to TECH2

- Two halves: programming fundamentals with Isabel (weeks 34–39), then working with data with Richard (weeks 40–45)
- One lecture and one workshop every week — you write code in both
- No previous programming experience assumed — we start from nothing
- All material is published on GitHub and linked from Canvas

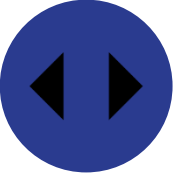
Goal for today: leave knowing what the course covers, how it runs, and what you need working before Friday.



Who Teaches What

- 1 Weeks 34–39 — Isabel Hovdahl: Python basics, decisions, loops, functions, packages
 - 2 Weeks 40–45 — Richard Foltyn: GitHub, NumPy, pandas, plotting, data wrangling, AI
- Part 1 teaches you to write Python. Part 2 teaches you to do something useful with it.
 - Same course, same tools, same habits — the second half builds directly on the first.

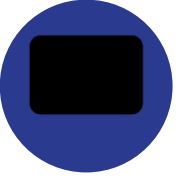
Questions: isabel.hovdahl@nhh.no · richard.foltyn@nhh.no



Why Python?

- Free and open source — no licence, runs on any machine you will ever own
- Readable enough to learn quickly, powerful enough for serious work
- General purpose: data analysis, visualization, automation, web, machine learning
- The most widely used programming language today, and the most used on GitHub

***Live rankings:** tiobe.com/tiobe-index · Fun fact: named after Monty Python, not the snake.*



Where TECH2 Fits in BEDS

Course	Semester	What you use Python for
TECH2 — this course	1	Programming, data analysis, visualization
TECH3: Applied Statistics	2	Statistical analysis
TECH4: Mathematics for Decisions	3	Optimizers and solvers
TECH5: Programming and Application Design	5	Object-oriented programming
TECH6: Data Science and Econometrics	5	Machine learning

This is a foundation course: everything you learn here you will use throughout the program.



How the Course Runs

- 1 Lecture — Wednesdays 08:15–10:00, aud. B: new ideas, worked examples, live coding
- 2 Workshop — Fridays 12:15–14:00, aud. B: you write the code, we help when you get stuck
- 3 Three mandatory assignments in weeks 35, 39 and 43 — all three must pass
- 4 Bring your laptop to both — you learn this by typing, not by watching

Learning happens in the classroom. Lectures and workshops are both hands-on, so participation is important.



Course Schedule

Week	Topic	Lecturer	Note
34	Course intro & Getting started with VS Code	Isabel	
35	Python basics & Version control with git	Isabel	Assignment 1
36	Decisions	Isabel	
37	Loops	Isabel	
38	Functions	Isabel	
39	Python packages & Virtual environments	Isabel	Assignment 2
40	GitHub & NumPy	Richard	
41	Intro to pandas	Richard	
42	Plotting	Richard	
43	Grouping and aggregation	Richard	Assignment 3
44	Concatenating and merging	Richard	
45	AI week	Richard	



What You Need Installed

- 1 VS Code — the editor you will write all your code in, from Friday onwards
- 2 Git — version control, used from week 35 and every week after that
- 3 Miniforge — Python itself, plus conda for managing packages
 - Already have Python installed some other way? Install Miniforge anyway — the course assumes it
 - Guides and a two-minute verification checklist: github.com/isabelhovdahl/TECH2 — see guides/

If something will not install, tell us today. We have set aside time in this lecture to fix it.



Two Kinds of Python File

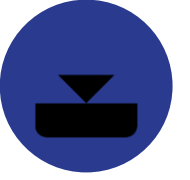
.py — Python script

- Plain text file of Python code
- Runs top to bottom, every time
- For programs and reusable code
- Mainly used in workshops

.ipynb — Jupyter notebook

- Code, output and notes in one document
- Runs cell by cell, in any order
- For exploring and explaining
- What the lecture material is written in

You will use both, and you open both in VS Code — you do not need a separate program for notebooks.



Where the Material Lives

- There is no book. Everything is on GitHub, free, and always the current version.
- Part 1: github.com/isabelhovdahl/TECH2
- Part 2: github.com/richardfoltyn/TECH2-H26
- On Canvas go to Modules — every week links to that week's material
- You do not need a GitHub account to read or download anything in part 1

If you can see it in Modules, it is ready. Material that is not linked yet may still be changing — do not work ahead from GitHub.

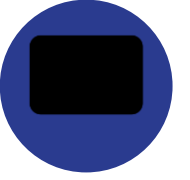


Assignments & Assessment

What	Weight	When
Three mandatory assignments — individual	Must pass all three	Weeks 35, 39 and 43
Digital school exam, 3 hours	70%	16 December
Term paper, groups of 2–3, 3 days	30%	23–25 November

- You must pass all three assignments to qualify for the exam.
- Assignments are handed in on Canvas — details are announced there for each one.

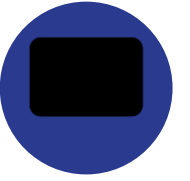
Keep up week by week. Each week builds directly on the last so gaps compound. Steady weekly work makes the assignments (and the course) easier.



Using AI in This Course

-  Ask what an error message means, or for a hint when you are properly stuck
-  Ask it to explain code you have already written yourself
-  Ask it for the solution, or accept code you cannot explain line by line
-  Use it to keep up rather than learn — the gap only shows up in the exam, when you are on your own
- Where AI contributes to work you hand in, say so. Each assignment tells you how.

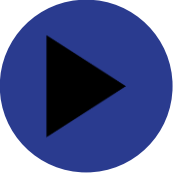
Use AI to understand things, not to produce things. Full guidelines: guides/ai-guidelines.md



Meet TECHi

- Built for this course — it knows the lecture material, the exercises, and where we are in the syllabus
- Privacy-friendly — your conversations are not sent to a commercial AI provider
- Socratic by design — it will not hand you the code. It asks questions and gives hints, so the thinking stays yours
- Find it on Canvas under Start here — nothing to install, no account to create
- You may still use other AI tools. The same guidelines apply, and so does declaring your use

It will not give you the answer, and that is the point. Try it on something you are genuinely stuck on — that is when it is most useful.



Before Friday

- 1 Check that VS Code, Git and Miniforge all work — the checklist takes two minutes
- 2 Open Canvas, go to Modules, and find this week's links
- 3 Read the AI guidelines — five minutes, and it will change how you work all semester
- 4 Bring your laptop to the workshop: Friday 12:15, aud. B

Nothing to prepare beyond this. Turn up on Friday and we start writing code.



See You on Friday

- Lectures Wednesdays 08:15 · Workshops Fridays 12:15 · aud. B
- All material on GitHub, linked from Canvas — Modules
- Install VS Code, Git and Miniforge before the workshop
- isabel.hovdahl@nhh.no · richard.foltyn@nhh.no

Friday: Getting Started with VS Code — your first Python script and your first notebook.